

Adaptive equidistant robot path planning on complex freeform surface based on 3D point cloud and least-squares conformal mapping

Huimin Shen, Kangming Liu, Shuangyuan Wang



PII: S1000-9345(26)00104-5

DOI: <https://doi.org/10.1016/j.cjme.2026.100315>

Reference: CJME100315

To appear in: *Chinese Journal of Mechanical Engineering*

Received date: 17 January 2025

Revised date: 21 September 2025

Accepted date: 24 April 2026

Please cite this article as: Huimin Shen, Kangming Liu and Shuangyuan Wang, Adaptive equidistant robot path planning on complex freeform surface based on 3D point cloud and least-squares conformal mapping, *Chinese Journal of Mechanical Engineering*, (2026) doi:<https://doi.org/10.1016/j.cjme.2026.100315>

This is a PDF of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability. This version will undergo additional copyediting, typesetting and review before it is published in its final form. As such, this version is no longer the Accepted Manuscript, but it is not yet the definitive Version of Record; we are providing this early version to give early visibility of the article. Please note that Elsevier's sharing policy for the Published Journal Article applies to this version, see: <https://www.elsevier.com/about/policies-and-standards/sharing#4-published-journal-article>. Please also note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2026 Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd.

RESEARCH ARTICLE

Adaptive equidistant robot path planning on complex freeform surface based on 3D point cloud and least-squares conformal mapping**Huimin Shen^{a,*}, Kangming Liu^a, Shuangyuan Wang^a**^a School of Mechanical Engineering, University of Shanghai for Science and Technology, Shanghai 200093, China*Corresponding author at University of Shanghai for Science and Technology, Shanghai 200093, China.
E-mail address: hmshen@usst.edu.cn

Peer review under the responsibility of Editorial Board of Chinese Journal of Mechanical Engineering.

Abstract

In complex freeform surface machining, such as aircraft skin grinding and engine blade processing, equidistant robot path planning is challenged by low efficiency and strong curvature influence on path uniformity. To address this, an adaptive equidistant path planning method based on 3D point cloud data and least-squares conformal mapping (LSCM) is proposed. The point cloud of the workpiece is first reconstructed into triangular meshes using greedy triangulation, and then the 3D path planning problem is reduced to a 2D plane through LSCM. Equidistant paths are generated on the plane and inversely mapped to 3D space via the area coordinate method, ensuring consistent spacing along complex curved surfaces. A complete robot path planning system integrating online and offline programming was developed and validated on an ABB industrial robot with a designed 3D-printed freeform surface. Experimental results show that the proposed method improves path isometry and normal accuracy, effectively enhancing machining efficiency and surface quality.

Keywords: Robot path planning, 3D point clouds, Least-squares conformal mapping, Complex curvature, Equidistant path planning

1 Introduction

In critical processes including grinding [1, 2], polishing [3-5], welding [6-8], and spraying [9, 10], interval path planning—such as robot work paths—is fundamental to achieving high processing efficiency, quality, and consistency. In the realm of intelligent manufacturing, particularly in aerospace applications such as engine blade machining and aircraft skin grinding, the challenges are compounded by intricate free-form surfaces and non-standard workpieces. Robotic operations in these scenarios are often hindered by low processing efficiency and reduced path planning accuracy in areas of high curvature. Taking the grinding of complex workpieces as an example [11], such as wind turbine blades, aero-engine blades or aircraft skins, often exhibit defects like burrs, weld residuals, or substandard surface roughness. These issues can adversely affect paint adhesion and aerodynamic performance, potentially leading to scrap or premature failure [12].

In order to improve the machining efficiency and quality, researchers have started to explore new methods for robot path planning [13]. Xu et al. [14] proposed a multi-objective optimization method for path planning of rigid-flexible coupled spraying robots, which integrated the spraying process constraints into the optimization process and effectively improved the spraying quality. In addition, Jiao et al. [15] investigated variable stiffness identification and configuration optimization of industrial robots in machining tasks to improve the positional accuracy and machining quality of the robots. These studies provide new methods for robot-assisted machining, which are crucial for improving machining consistency, reducing machining errors, and enhancing machining efficiency.

Existing path planning methods primarily rely on two approaches: Manual instruction for online programming and computer-aided design (CAD) models for offline programming. Manual teaching methods are not only time-intensive but also highly dependent on operator experience. For workpieces with complex curved geometries in aerospace and other industries, manual teaching proves inadequate for the demands of multi-variety, small-batch production lines [16]. Moreover, in scenarios involving chemical hazards such as spraying, prolonged tool manipulation and precise control pose a significant threat to operator health [17]. Path planning based on CAD models is also utilized for robot machining path planning [18]. However, this approach has limitations. The creation of CAD models relies on precise dimensions and shape information of the workpiece, but in actual machining, installation errors and deformations often cause the workpiece to not fully match the CAD model, which affects the accuracy of path planning and subsequently impacts the quality and efficiency of machining [19]. Jahanpour et al. [20] proposed a CAD-based smooth path

planning scheme using parametric non-uniform rational B-spline (NURBS) curves for parallel machining robots. Although this method can generate smooth trajectories, it may require additional correction steps to ensure path accuracy when dealing with actual installation errors and deformations of the workpiece.

In cases where CAD models are unavailable or the workpieces are easily deformable during clamping and machining, alternative approaches based on measurement point cloud data have been investigated [21]. Bastidad et al. [16] addressed the challenges of robotic spraying operations by extracting surface features from measured point cloud data and employing a "wrapping" algorithm to generate trajectory paths covering the surface. Similarly, Zhang et al. [22] developed a path trajectory planning method involving point cloud slicing and discretized fitting of segmented paths using B-spline curve fitting and curvature discretization technique.

However, conventional path planning methods face limitations. When the normal direction of a planned surface area is nearly parallel to the normal direction of the driving plane, especially in high-curvature regions, interval-based path planning often results in reduced spacing, leading to redundant or overlapping machining paths. This redundancy decreases machining efficiency. To address these challenges, Zhu et al. [23] proposed a method for obtaining path by fitting the surface of the model to be processed. Yang et al. [24] introduced a polishing path planning technology for small aerospace blades that combined reverse engineering with finite element mesh technology. Their approach reconstructed 3D surface models from point cloud data, mapping grid nodes to finite element meshes to derive polishing paths through grid point sequencing.

A highly versatile path planning method with simplified processes is critical for ensuring consistency in robot-assisted machining, minimizing errors, and enhancing efficiency [25]. Mathematical tools and methodologies have proven effective for path planning on complex surfaces. In 2002, Lévy et al. [26] proposed least-squares conformal mapping (LSCM) to map 3D surfaces to 2D parameter spaces. By minimizing angular distortion and uniformly scaling to maintain consistent triangle side ratios, LSCM effectively handles complex geometric transformations while preserving the accuracy of geometric features. The linearity of LSCM further improves the efficiency and feasibility of parameterization processes [27]. Initially developed for texture mapping, LSCM has evolved into a pivotal technique in computer graphics, with its applications extending into numerous related fields, cementing its importance in the study of complex geometric transformations.

In this paper, an adaptive equidistant path planning method based on 3D point cloud data and LSCM is proposed for complex freeform surface machining with various curvatures. Through point cloud reconstruction and LSCM technology, the 3D path planning problem is reduced to a 2D plane for processing, so that the path spacing can adapt to the local geometric features of the surface. By preprocessing the point cloud data, the complex surface triangular mesh is constructed, and is further expanded by LSCM. This method ensures the accurate mapping relationship from the 3D physical space to the 2D parameter space while maintaining the continuity of the grid structure. Then, the path is planned at equal intervals on the generated 2D planar grid. Finally, the 2D path points are inversely mapped to the 3D space by the area coordinate method, so as to generate the robot operating path that adapts to different curvature surface geometries.

The rest of this paper is structured as follows. Section 2 introduces the uniformly spaced path planning method, including the detailed method flow from 3D surface modeling to robot communication. Section 3 describes the experimental platform and experimental scheme. Section 4 shows the experimental results and analysis. Section 5 summarizes the research contributions and future research directions.

2 Methodology

The technical route of the proposed adaptive equidistant robot path planning on complex freeform surface is shown in Fig. 1.

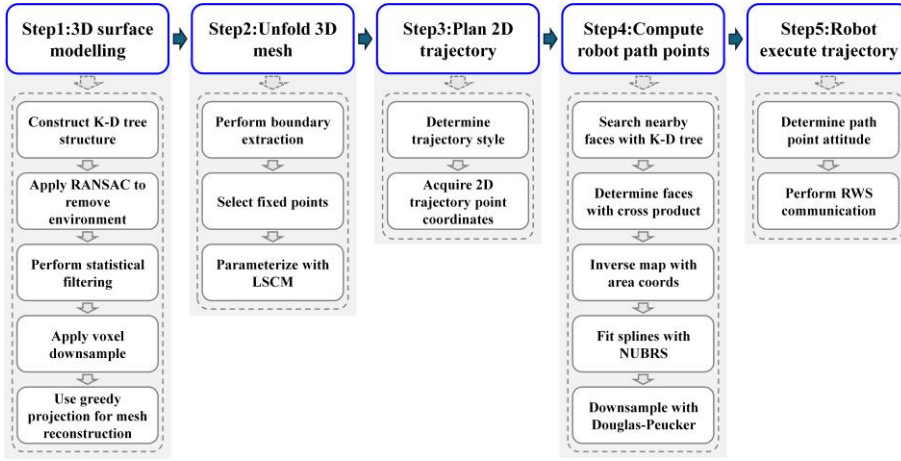


Fig. 1 Technology roadmap

- **Step 1:** Utilize the greedy triangle method to generate a triangular mesh from the pre-processed point cloud data, achieving 3D mesh reconstruction for the complex surface to be processed.
- **Step 2:** Employ LSCM to unfold the 3D mesh into a 2D planar grid, performing LSCM parameterization.
- **Step 3:** Design the path at equal intervals on the unfolded 2D plane to acquire the coordinate information of the 2D path points.
- **Step 4:** Map the equidistantly planned path points from the 2D plane back to the 3D space using the area coordinate method, generating an equidistant operational path for the robotic system. This path is capable of adapting to complex surface curvatures.
- **Step 5:** The robotic system executes the processing task following the planned path.

2.1 3D Mesh Surface Modeling

3D surface mesh reconstruction is a critical step in achieving 3D surface unfolding. As illustrated in Fig. 1, to enhance the processing efficiency of high-density point cloud data, a K-D tree structure is employed for efficient indexing. The RANSAC algorithm, combined with statistical filtering, is utilized to eliminate background points and noise, ensuring both the accuracy and cleanliness of the reconstructed 3D model. Subsequently, voxel downsampling technology is applied to simplify the point cloud data while preserving the geometric features of the complex surface intended for planning. Finally, the greedy triangle mesh construction method is used to generate a 3D triangular mesh model of the workpiece based on the simplified point cloud data.

To achieve efficient neighborhood retrieval in 3D point cloud preprocessing, this study employs the K-D tree data structure to index the collected point clouds. To further enhance the processing efficiency of high-density point cloud data, voxel downsampling is applied, effectively reducing the number of points in the point cloud. This approach not only preserves the overall structure and key geometric features of the point cloud but also significantly decreases the computational complexity of subsequent processing steps.

To address the inherent disorder and discreteness of point cloud data, the greedy triangular mesh construction method is utilized to generate continuous triangular gridded surfaces. The local optimization characteristics of the greedy algorithm allow it to construct triangular grids that closely align with local regions of the point cloud, without requiring global optimality [28]. Furthermore, Delaunay triangulation is employed to generate triangles that satisfy the "no point inside the circumcircle" property, maximizing the minimum angles of the triangles. This approach prevents the formation of excessively narrow and elongated triangles, thereby ensuring the geometric stability and quality of the resulting mesh.

2.2 2D Mesh Unfolding Based on LSCM

Due to the complexity of path planning for 3D surfaces, conformal mapping technology based on the LSCM is employed to map the 3D surface onto a 2D plane. During the dimensionality reduction process, the geometric shape characteristics of the 3D surface are preserved, ensuring an accurate representation of the 3D structure within the 2D space.

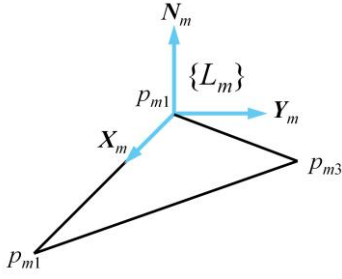


Fig. 2 Local coordinate system of a triangle

As shown in Fig. 2, a local coordinate system $\{L_m\}$ is constructed on each triangle of the triangular mesh to convert 3D coordinates to local 2D coordinates. For each spatial triangle $s_m = \triangle p_{m1}p_{m2}p_{m3}$, the local axes X_m and Y_m are defined, along with the normal vector N_m . The local coordinate system is constructed as Eq. (1).

$$\begin{cases} X_m = \frac{p_{m2} - p_{m1}}{\|p_{m2} - p_{m1}\|}, \\ N_m = \frac{X_m \times \{p_{m3} - p_{m1}\}}{\|X_m \times \{p_{m3} - p_{m1}\}\|}, \\ Y_m = N_m \times X_m \end{cases} \quad (1)$$

where p_{m1} and p_{m2} are the two vertices of the triangular element s_m .

Then, complex variable $s=x+iy$ is utilized to represent the local coordinate system on a single triangle, where (x, y) is the local coordinate. Similarly, for uv coordinates on the parameter field, complex variable $t=u+iv$ is taken for definition. In the reconstructed discrete 3D grid $\mathcal{S}=\{s_m\}_{m=1, \dots, n}$, where n signifies the number of triangles of the reconstructed surface mesh. A smooth mapping complex function ψ that can realize $\mathcal{S}$ mapping to the (u, v) space, $t_m=\psi(s_m)$ (where $t_m=u_m+iv_m$), should be figured out. In the mapping process, to remain the local angles unchanged and achieve conformal, the mapping is required to satisfy the Cauchy-Riemann equation and constraint conditions [29]:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}, \quad (2)$$

$$\frac{\partial \psi}{\partial x} + i \frac{\partial \psi}{\partial y} = 0. \quad (3)$$

However, due to the practical constraints, it may not be feasible to fully satisfy the conformal condition. Therefore, the conformal mapping is approximated by minimizing a least-squares based objective function according to Ref. [30]. Then, the objective function ELSCM can be formulated by:

$$\begin{aligned} E_{LSCM} &= \sum_{s_m \in \mathcal{S}} \int_{s_m} \left| \frac{\partial \psi}{\partial x} + i \frac{\partial \psi}{\partial y} \right|^2 dA \\ &= \sum_{m=1}^n T_m \left\| \mathbf{M}_{T_m} \begin{pmatrix} v_{m1} \\ v_{m2} \\ v_{m3} \end{pmatrix} - \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \mathbf{M}_T \begin{pmatrix} u_{m1} \\ u_{m2} \\ u_{m3} \end{pmatrix} \right\|^2, \end{aligned} \quad (4)$$

$$\mathbf{M}_{T_m} = \frac{1}{2T_m} \begin{pmatrix} y_{m2} - y_{m3} & y_{m3} - y_{m1} & y_{m1} - y_{m2} \\ x_{m3} - x_{m2} & x_{m1} - x_{m3} & x_{m2} - x_{m1} \end{pmatrix}, \quad (5)$$

where T_m denotes the area of the triangular mesh s_m defined by the points (p_{m1}, p_{m2}, p_{m3}) and $\mathbf{M}_{T_m}$ denotes the matrix associated with T_m .

By minimizing the objective function in Eq. (4), the plane coordinates of the 3D triangular mesh in the parameter space (u, v) can be obtained. The 3D mesh can be described in the 2D parameter plane (u, v) . The specific algorithmic flow for the implementation of the LSCM parameterization is shown in Table 1.

Table 1 Algorithmic flow of LSCM parameterization

Algorithm 1. LSCM Parameterization	
1	Procedure LSCM Parameterization (Mesh)
2	Query the index of vertices in input mesh $\mathcal{S}$ to obtain each triangle T_m

3	Compute the area of every triangle T_m and matrix M_{T_m}
4	Compute boundary vertices P_{BOU} and interior vertices P_{INT} in S
5	Construct the energy function E_{LSCM} using Eq. (4)
6	Select two locked vertices in boundary vertices P_{BOU}
7	Compute the boundary vertices P_{BOU} as constraint points
8	Solve a linear system of equations to obtain the planar coordinates (u, v) in parametric space
9	Return the parameterized mesh in the 2D parametric plane
10	End procedure

2.3 Calculate Robot Path Points

2.3.1 3D inverse mapping

Then, the equidistant robot path can be planned directly on the expanded 2D plane according to the specific machine requirements and the geometric characteristics of the workpiece. The planned path can be inversely mapped into 3D space. To implement 3D surface path planning, it is essential to identify the triangular mesh on the 3D surface corresponding to the mapped path points on the unfolded 2D mesh. By utilizing the K-D tree structure of the original point cloud, the N nearest neighboring points for each path point on the unfolded 2D mesh are determined. These nearest points are then organized into a set of triangular meshes. The specific triangular mesh containing each 2D path point is identified using the vector cross-product method, along with the corresponding indices of these facets.

Subsequently, an inverse mapping from the 2D mesh to the 3D surface mesh is performed. This is achieved through the area coordinate method, enabling the retrieval of the 3D spatial position information corresponding to the path points on the 2D mesh.

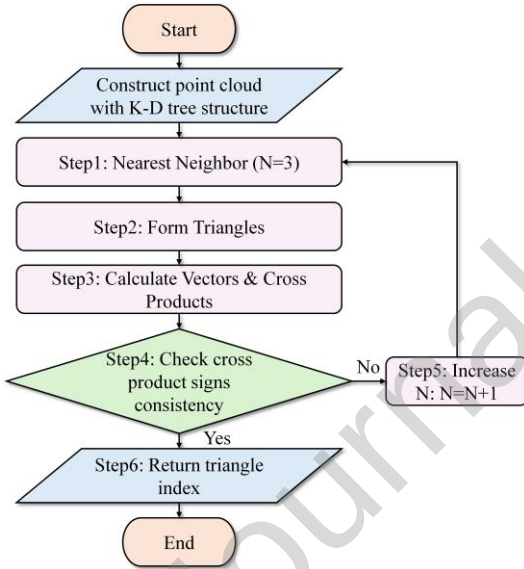


Fig. 3 Determining path points via K-D Tree and cross product

The flowchart for obtaining the 3D spatial position information corresponding to a 2D path point P is shown in Fig. 3.

- **Step 1:** Nearest-neighbor search for the current 2D path point P with initially selected $N=3$;
- **Step 2:** Formulation of all possible 2D triangles by the three of the searched nearest neighbor points in the 2D grid;
- **Step 3:** Calculation of the vectors from the current 2D path point P with respect to all the possible 2D grid triangle vertices A_m, B_m, C_m , termed PA_m, PB_m and PC_m and the cross-product of two vectors according to Eq. (6):

$$\text{Cross1} = PA_m \times PB_m$$

$$= (x_{A_m} - x_P)(y_{A_m} - y_P) - (y_{A_m} - y_P)(x_{B_m} - x_P),$$

$$\text{Cross2} = PB_m \times PC_m$$

$$= (x_{B_m} - x_P)(y_{C_m} - y_P) - (y_{A_m} - y_P)(x_{B_m} - x_P), \quad (6)$$

$$\text{Cross3} = PC_m \times PA_m$$

$$= (x_{C_m} - x_P)(y_{A_m} - y_P) - (y_{C_m} - y_P)(x_{A_m} - x_P).$$

- **Step 4:** If the computed signs of *Cross1*, *Cross2*, and *Cross3* are the same, it can be determined that the 2D point P is inside a 2D triangle. Then, return the index of the corresponding 2D triangular mesh.
- **Step 5:** If Step 4 cannot be satisfied, increase the number of the nearest-neighbor points N and repeat the former process until a 2D triangle containing the point is found.
- **Step 6:** Once a 2D triangle containing point P is confirmed, return the index of the 2D triangular mesh.

While ensuring the positioning accuracy of path points, the above dynamic nearest-neighbor points number adjustment method can effectively reduce the computational load, thereby providing an efficient and reliable foundation for subsequent 3D inverse mapping.

Following the determination of the 2D path point P as being inside a specific triangle $\Delta A_m B_m C_m$ of the parameter field, the area coordinate method is then utilized to accurately calculate the corresponding 3D position on the triangular mesh. This step is essential for translating the 2D planning into the 3D operational space where the robot will execute the task. The position of point P within the triangle is defined by the area coordinates $(\lambda_{m0}, \lambda_{m1}, \lambda_{m2})$, which are calculated based on the areas of the sub-triangles formed with P as a vertex:

$$\lambda_{mj} = \frac{D_{mj}}{D_m}, (j = 0, 1, 2), \quad (7)$$

where D_{m0} , D_{m1} , D_{m2} , D_m represent the area of $\Delta B_m C_m P$, $\Delta C_m A_m P$, $\Delta A_m B_m P$, $\Delta A_m B_m C_m$ respectively; obviously, $\lambda_{m0} + \lambda_{m1} + \lambda_{m2} = 1$. By the definition of the area coordinate system, point P can be represented linearly by λ_{m0} , λ_{m1} , λ_{m2} :

$$P = \lambda_{m0} A_m + \lambda_{m1} B_m + \lambda_{m2} C_m. \quad (8)$$

Given the spatial triangle $\Delta a_m b_m c_m$ that corresponds to the 2D triangle $\Delta A_m B_m C_m$, the 3D coordinates of point P in the spatial triangle can be derived using the following formula:

$$Q = \lambda_{m0} a_m + \lambda_{m1} b_m + \lambda_{m2} c_m. \quad (9)$$

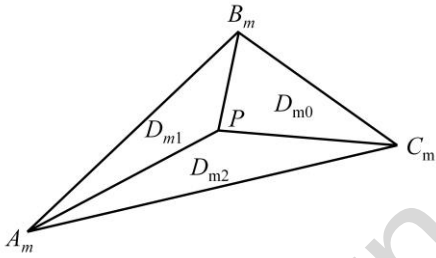


Fig. 4 Area coordinate of a specific triangle $\Delta A_m B_m C_m$

This process, facilitated by the area coordinate method, enables the rapid determination of the 3D spatial points corresponding to any 2D path point, ensuring a precise 3D path alignment of the robot with the planned 2D path. Fig. 4 illustrates the visual representation of the area coordinate system.

2.3.2 Track point fitting and downsampling

Based on the 3D inverse mapping to obtain the robot path points Q in 3D space, these points need to be refined to ensure the smoothness and efficiency of the robot path planning. This is crucial for enhancing the operational performance of robot and ensuring the path accuracy.

To address potential issues with the distribution and density of these points, trajectory point fitting and downsampling is applied to each trajectory path obtained through inverse mapping. Utilizing Non-Uniform Rational B-Spline (NURBS) curve fitting, the control vertices and weighting factors according to the specific requirements of the robot path and the geometric characteristics of the workpiece can be precisely adjusted. This method enables the NURBS technique to accurately fit curves of various shapes, enhancing the precision of free-form surface contour fitting. Each trajectory path is represented by a segmented rational polynomial vector function, as shown in Eq (10).

$$p(u) = \frac{\sum_{i=0}^n \omega_i d_i N_{i,k}(u)}{\sum_{i=0}^n \omega_i N_{i,k}(u)}, \quad (10)$$

where ω_i is the weight factor related to the control vertex d_i ($i=0,1,\dots,n$); $N_{i,k}(u)$ represents the B-spline basis function of degree k , which can be deduced from Eq. (11):

$$\begin{cases} N_{i,0}(u) = \begin{cases} 1, u \in [u_i, u_{i+1}], \\ 0, \text{other}, \end{cases} \\ N_{i,k}(u) = \frac{u - u_i}{u_{i+k+1} - u_i} N_{i,k-1}(u) + \frac{u_{i+k} - u}{u_{i+k} - u_{i+1}} N_{i+1,k-1}(u), \\ \text{rule: } \frac{0}{0} = 0. \end{cases} \quad (11)$$

NURBS curves are discretized into a series of evaluation points for downsampling using the Douglas-Peucker algorithm. The evaluation point $C(u_i)$ is sampled in the parameter interval $[0, 1]$. The specific formula is as follows:

$$u_v = \frac{v}{l-1}, \quad (12)$$

$$C(u_v) = \frac{\sum_{i=0}^n N_{i,p}(u_v) \cdot P_i \cdot \omega_i}{\sum_{i=0}^n N_{i,p}(u_v) \cdot \omega_i}, \quad (13)$$

where u_v denotes the sampling point for parameter u and v denotes the number of evaluation points.

To further refine the point spacing on the path and ensure a balance between path detail and computational efficiency, the Douglas-Peucker method is applied to thin out the evaluation points. The principle of the Douglas-Peucker algorithm is shown in Fig. 5: This is done by calculating point C , which represents the maximum deviation value d between line segment AB and its approximated curve, and comparing it to a set threshold r . If d is less than r , AB is considered a new insertion point for the curve segment; otherwise, when d is larger than r , point C is used as the new insertion point, and this process is repeated until the maximum deviation value for each curve segment is less than r .

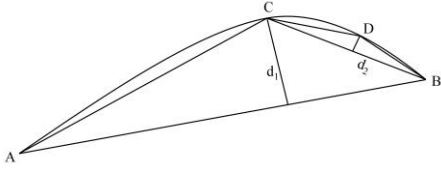


Fig. 5 Douglas-Peucker algorithm

3 Experimental Setup and Procedure

3.1 Experimental Configuration

In order to demonstrate the feasibility of the proposed path planning method, a robot path execution system is developed, as depicted in Fig. 6. This system includes a 3D vision system and a robot system. The 3D vision system encompasses a 3D binocular structured light camera (Mech-Mind™, Mech-Eye PRO S) and a personal computer (PC: CPU i7-13800H, 32 GB memory). The detailed parameters of the 3D camera are presented in Table 2. The path execution system comprises an industrial robot (ABB™ IRB1200) and a self-designed fixture for a marker pen. The fixture is fixed on the end effector of the robot, while the 3D camera is fixedly installed along the positive X_0 -axis direction of the robot base, adopting the configuration of eye outside the hand. The end of the marker pen is connected to a spring, allowing the marker pen to extend and retract along the Z_{TCP} -axis direction of the default TCP of the robotic arm. Thus, the planned path can be vividly drawn on paper for quantitative evaluation and analysis.

The experimental workpiece was modeled in SolidWorks by drawing a spline curve and extruding it to form a freeform surface. To characterize its geometric complexity, the curvature of the surface was analyzed using the SolidWorks measurement tool. The results show that the curvature ranges from 0.000998 mm^{-1} to 0.012663 mm^{-1} , indicating that the workpiece contains both nearly flat regions and highly curved regions. This variability reflects the freeform characteristics of the surface and makes it a suitable model for evaluating the adaptability of the proposed path planning method.

When planning the path, the workpiece should first be positioned beneath the 3D camera to ensure complete coverage within its field of view. The 3D camera is then used to capture the point cloud data of the workpiece, which is transmitted to the computer via the Ethernet interface. Finally, the point cloud data is processed on the PC to plan the robot's paths, and the robot is subsequently controlled to execute the planned path.

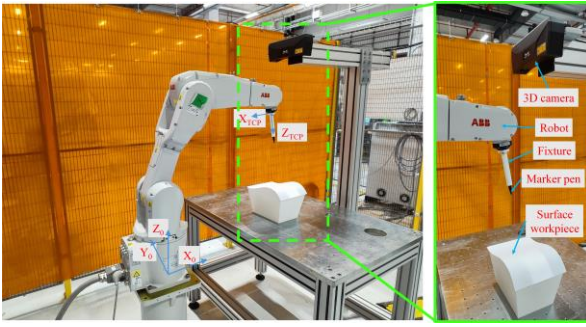


Fig. 6 Robot path execution system

Table 2 Parameters of Mech-Eye PRO S

Parametric	Value
Recommended working distance (mm)	500—1000
Near field of view	370 mm×240 mm @ 0.5 m
Far field of view	800 mm×450 mm @ 1.0 m
Z-axis single point repeatability σ	0.05 mm @ 1.0 m
VDI/VDE measurement accuracy	0.1 mm @ 1.0 m
Communication interface	Gigabit Ethernet

3.2 Experimental Verification

To validate the efficacy of the proposed path planning approach based on 3D point cloud data and LSCM, a comprehensive validation experiment was designed, consisting of several key steps.

- Firstly, a high-precision 3D camera (Mech-Mind™, Mech-Eye PRO S) was employed to capture the experimental workpiece to acquire its surface point cloud data, and the data was preprocessed through procedures such as establishing a K-D tree structure, extracting the workpiece by RANSAC, statistical filtering, and voxel downsampling.
- Subsequently, the greedy projection triangulation method was utilized to reconstruct the 3D surface model of the workpiece. Next, the LSCM method was applied to flatten the 3D surface onto a 2D plane, and parallel and equidistant straight lines were planned on the plane. The path points in the 3D space were obtained via inverse mapping, and then these path points were fitted and downsampled.
- Finally, the planned path was imported into the robot control system.

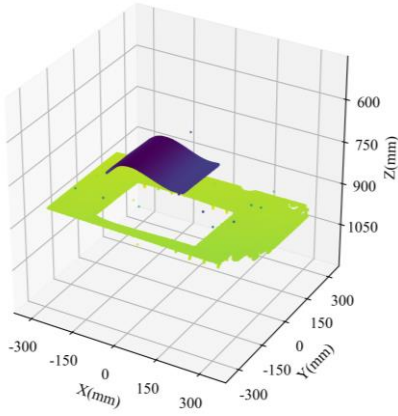
To validate the path adaptability and distribution rationality, a pen was installed at the end of the robot, and path were drawn on the white paper covering the workpiece. Once the white paper was flattened, the path lines were extracted through image processing technology, and the spacings were analyzed. By comparing the spacings and distributions of the path lines before and after the experiment, it was verified that the path spacings could adapt to the local geometric shape of the curved surface, and the path distribution was more reasonable. Additionally, by planning concentric circles and equidistant spiral lines, the robustness of this method was further verified, ensuring consistent path spacings and distributions in different path shapes and directions, thereby demonstrating the applicability and reliability of this path planning method in various application scenarios.

4 Results and Analysis

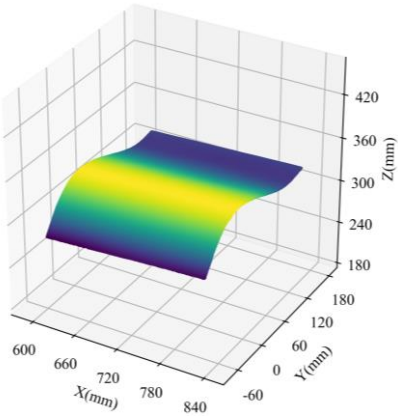
4.1 Experimental Data

4.1.1 Point cloud preprocessing

The original point cloud encompasses background points within the non-workpiece area. Firstly, the RANSAC algorithm is employed to eliminate the background points and acquire the workpiece point cloud. The separation effect is depicted in **Error! Reference source not found.**



(a) Original point cloud obtained



(b) Separated workpiece point cloud

Fig. 7 Comparison of environmental information before and after separation

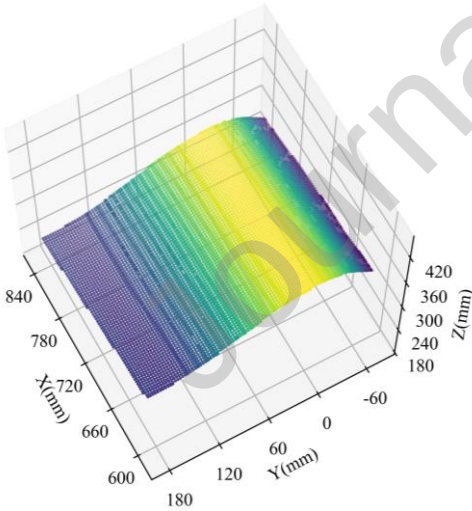


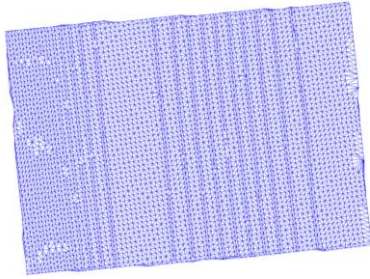
Fig. 8 Point clouds after voxel downsampling

To reduce computational complexity and accelerate processing speed while preserving key shape features of the workpiece, voxel downsampling was applied. **Error! Reference source not found.** illustrates the point clouds after voxel downsampling. By setting a voxel grid side length of 3 mm, we reduced the number of points from 600000 in the original point cloud to 9000 after voxel filtering, ensuring that the essential shape characteristics were maintained despite the significant data reduction.

According to the greedy projection triangulation method in Section 2.1, the triangulation meshes of different local areas of the downsampled workpiece model can be effectively spliced, and finally the whole triangulation mesh surface can be formed. The reconstructed surface model is shown in **Error! Reference source not found.**(a).



(a) Reconstructed surface model



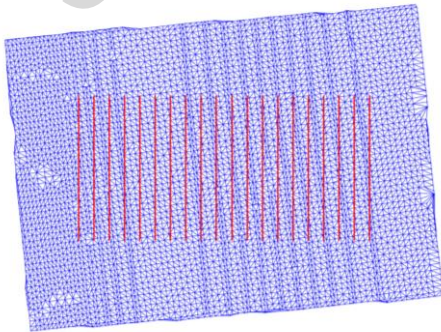
(b) 2D grid model

Fig. 9 3D Surface Reconstruction and 2D Mesh Mapping Using LSCM

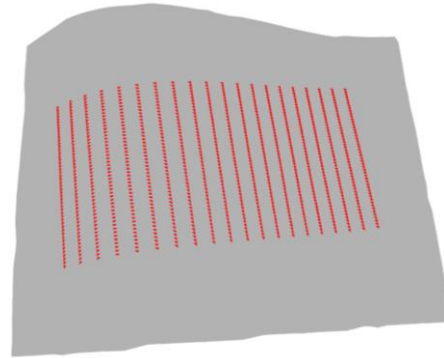
Error! Reference source not found.(b) displays the results of applying LSCM to the 3D surface model reconstructed in **Error! Reference source not found.**(a). This step is crucial as it demonstrates precisely how complex 3D geometries are accurately mapped onto a 2D plane, facilitating precise path planning. A key feature of LSCM parameterization is its ability to maintain the continuity of the triangular mesh while minimizing distortion energy during the transformation. This capability is essential for ensuring that the path planning process can accurately reflect the geometric characteristics of the 3D model in the 2D plane, thus providing a reliable basis for subsequent path planning steps.

4.1.2 Parallel line path planning

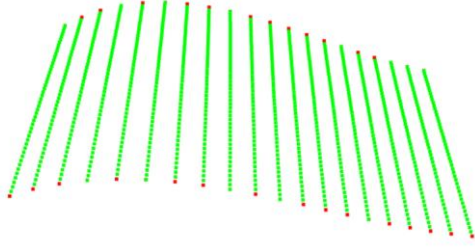
In this study, the validity of the path planning method is verified by drawing different numbers of path lines (20, 10, 5) in the same area. Fig. 7 shows the whole process of parallel line path planning: Fig. 7(a) shows the path planning of 20 parallel lines on a 2D plane, where the red line represents the dense path points of the planning. Fig. 7(b) shows the result of converting these 2D path points into 3D surfaces through the inverse mapping process, and the red dots mark the mapped path. In Fig. 7(c), the 3D path points after fitting and downsampling are shown in green, representing the optimized robot path points. Finally, Fig. 7(d) shows the simulation results in RobotStudio, which verify the accuracy and feasibility of path planning in the form of yellow dots.



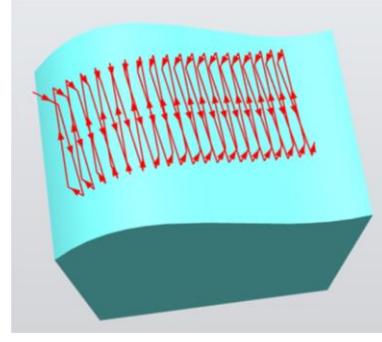
(a) 2D plane path planning



(b) 3D path points after inverse mapping



(c) 3D path points after fitting and downsampling
Fig. 7 Twenty straight line planning process



(d) RobotStudio simulation results

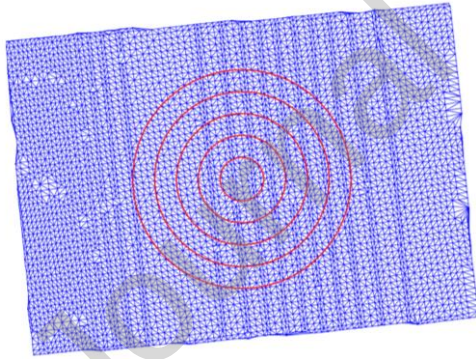
4.1.3 Equidistant concentric circles and equidistant spiral

Further, in order to comprehensively evaluate the robustness of the path planning method, the planning experiments of equidistant concentric circles and equidistant spiral are carried out. These experiments, also carried out on a 2D plane, involve mapping a 3D surface to a 2D plane and planning equidistant concentric circles and spiral path on that plane.

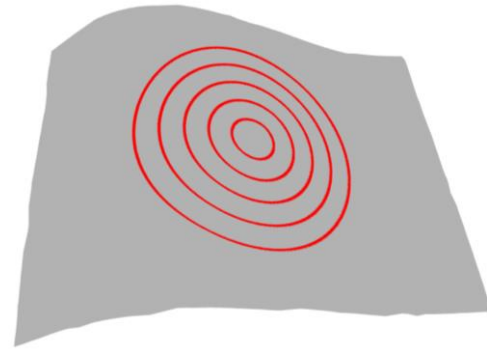
Fig. 8 shows the results of path planning for equidistant concentric circles. The circles of different radii in the figure represent path of equidistant concentric circles planned on a 2D plane. Through inverse mapping, these 2D circles are transformed into path points in 3D space and displayed as red dots. Similarly, Fig. 9 shows the results of equidistant spiral path planning.

4.2 Error Analysis and Rationality Verification

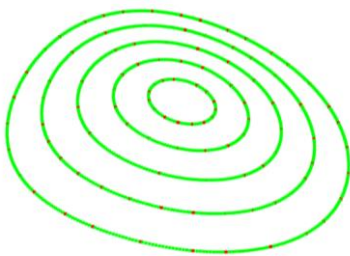
This section quantitatively analyzes the equidistance of the proposed path planning method in 3D space. Through a series of design experiments, we explore the isometry of different number of track lines, and use boxplot and standard deviation table to visually display the distance statistics results to evaluate the isometry of track lines.



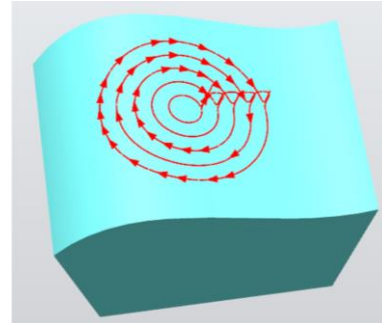
(a) 2D plane path planning



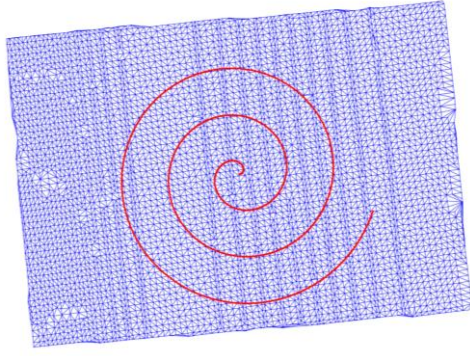
(b) 3D path points after inverse mapping



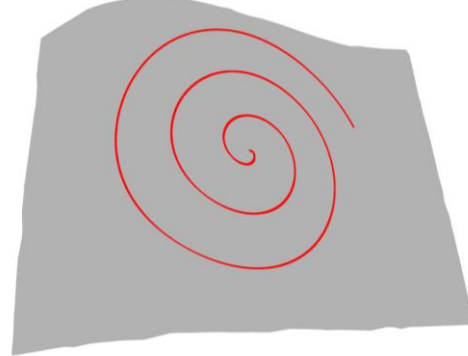
(c) 3D path points after fitting and downsampling
Fig. 8 Equidistant concentric circle planning process



(d) RobotStudio simulation results



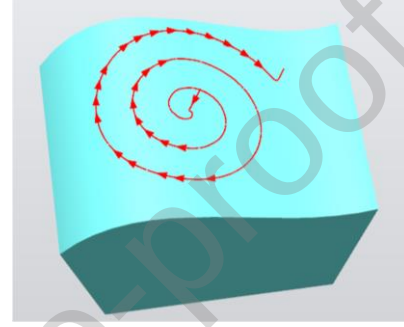
(a) 2D plane path planning



(b) 3D path points after inverse mapping



(c) 3D path points after fitting and downsampling



(d) RobotStudio simulation results

Fig. 9 Equidistant spiral planning process

In order to further visually display the experimental results, Fig. 10 provides the results of the experimental scan. These images show the traces left by the robot in accordance with the planned track line on the white paper, and the track line data is extracted by a calibrated Cognex™ acA1440-73gc industrial camera and image processing technology. Based on the camera calibration, the extracted pixel coordinates were converted into the world coordinate system, so that the spacing results are reported as absolute distances (mm). Image processing includes gray-scale, binarization, region-growing method to segment line segments, skeletonization and using RANSAC algorithm to fit straight line equations, and then calculate the minimum distance between each discrete point and adjacent trace lines. The experimental results are as follows.

In Fig. 11, the minimum distance distribution of different experimental groups (20, 15, 5 lines), denoted as d_{line} , was analyzed by boxplot, and the isometry and spacing uniformity of track lines were evaluated. The median absolute distance (mm) was similar across all experimental groups, indicating that the mean spacing remained consistent across conditions. The Interquartile Range (IQR), defined as the range between the first quartile (25th percentile) and the third quartile (75th percentile) of the data, is used to measure statistical dispersion and identify outliers. In our analysis, the IQR is small, indicating that there are no obvious outliers. The distribution patterns of the data are displayed, and the experimental results are stable. This suggests low variability and good consistency in the path planning results. The symmetry of the boxplot further verifies the consistency and reliability of the experimental method.

Table 3 lists the standard deviation of absolute distance (mm) in each experimental group, providing quantitative evidence of the consistency of track line spacing. For the 20-line group, the standard deviations of the three experiments were 0.080, 0.085, and 0.078, showing high consistency. As the number of trace lines decreases, the standard deviation increases slightly, but the overall level remains low, indicating that the method can maintain good isometry under different conditions.

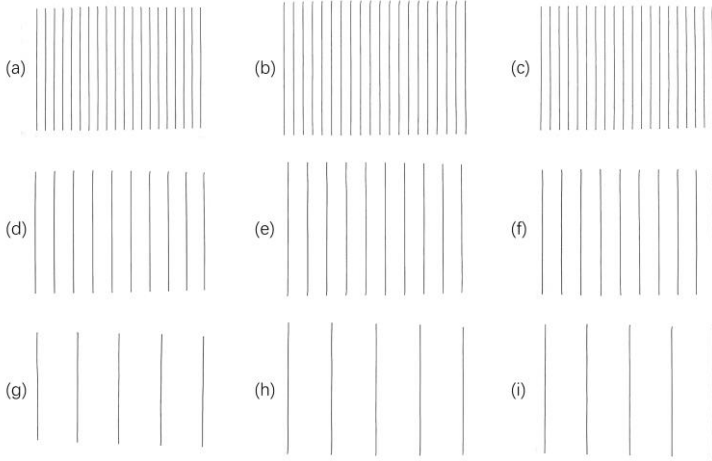


Fig. 10 Plot of experimental scanning results: (a)–(c) Results of 20 line, (d)–(f) Results of 10 line, (g)–(i) Results of 5 line.

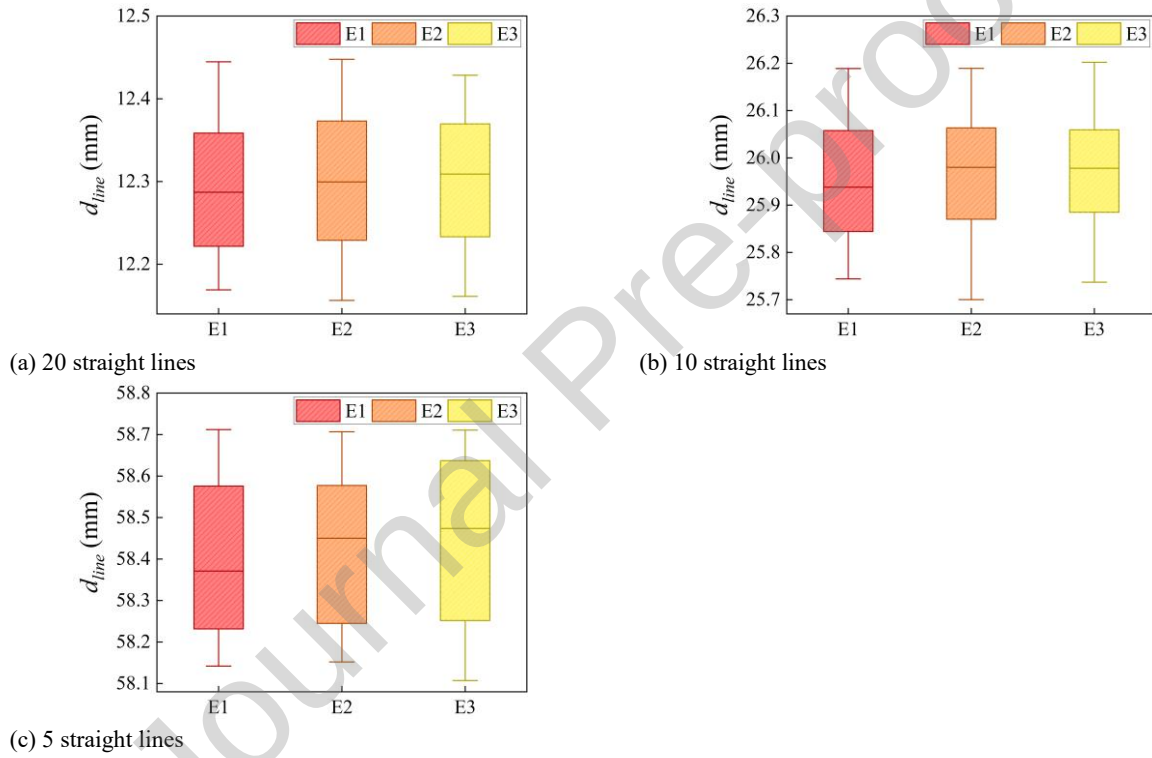


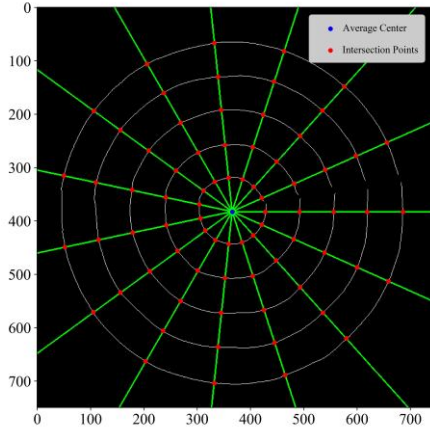
Fig. 11 Boxplots of minimum distances for each experimental group

Table 3 Standard deviation of path line spacing for different line counts

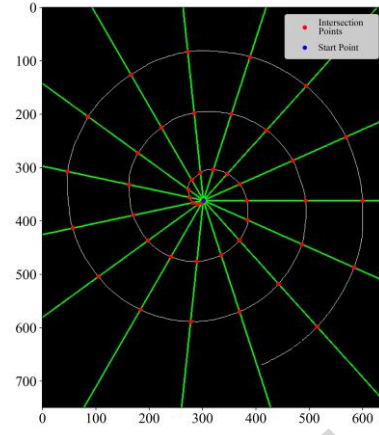
Number of path lines	Experiment 1 (E1) (mm)	Experiment 2 (E2) (mm)	Experiment 3 (E3) (mm)
20 lines	0.080	0.085	0.078
10 lines	0.128	0.127	0.115
5 lines	0.186	0.180	0.195

4.3 Robustness Verification

This section experimentally evaluates the robustness of the path planning approach to straight, concentric and helical paths. Using image processing techniques, as shown in Fig. 12, we recorded the pixel coordinates of the intersection points of 15 rays emanating from the path starting point with the path and calculated the spacing between adjacent intersection points. The box-and-line plot in Fig. 13 illustrates the distribution of absolute distances (mm) for the two path planning methods. The distances for the concentric circle path are denoted as d_{circle} , and the distances for the spiral path are denoted as d_{spiral} . Table 4 provides the standard deviation of the spacing for different path types.

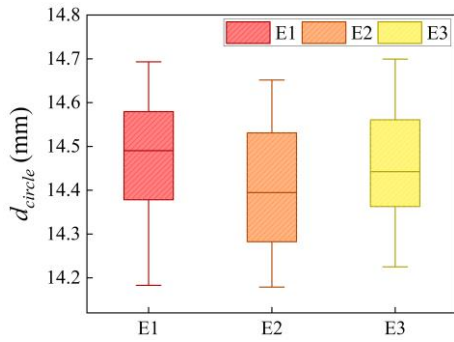


(a) Intersection Points and Lines from Average Center

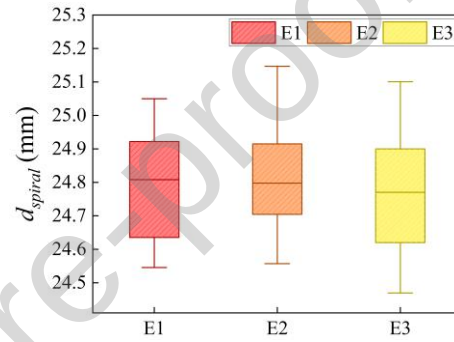


(b) Intersection Points and Lines from Start Point

Fig. 12 Intersection of equidistant concentric circles and equidistant spiral



(a) Equidistant concentric circles



(b) Equidistant spiral

Fig. 13 Distance in absolute units (mm) between adjacent intersections of equidistant spiral and equidistant concentric circles

Table 4 Standard deviation of different path track spacing

Path type	Straight lines			Circle	Spiral
	20	10	5		
Standard deviation (mm)	0.081	0.125	0.189	0.146	0.179

The experimental results, as depicted in Fig. 13 demonstrate that the absolute distances (mm) for equidistant concentric circle path planning are predominantly centered between 14.2 and 14.7 mm. In contrast, the absolute distances (mm) for equidistant spiral path are spread across a broader range, from 24.5 to 25.2 mm. This slightly wider distribution for both path types can be attributed to the non-rigid nature of the spring-loaded pen, which introduces a degree of variability in the curved path planning. However, the standard variance for each path type, as detailed in Table 4, remains relatively small, suggesting a high consistency in spacing.

Despite the slightly greater variability observed in both concentric circle and spiral path planning, the robustness of the proposed path planning algorithms is evident across different path shapes. The standard deviation for each path type, as shown in Table 4, is relatively low, indicating a good consistency in spacing. Specifically, for 20 straight lines, the standard deviation is 0.081 mm, for 10 straight lines it is 0.125 mm, and for 5 straight lines it is 0.189 mm. For circular path, the standard deviation is 0.146 mm, and for spiral path, it is 0.179 mm.

Although concentric circle and spiral path planning show slightly greater variability, the proposed path planning algorithms show good robustness for different path shapes.

4.4 Normal Accuracy Test of Path

To evaluate the normal accuracy of the planned path on freeform surfaces, a dedicated experiment was conducted. In the setup, the path marking tool at the robot end-effector was replaced with a digital dial gauge, and tool coordinate calibration was performed to align the gauge probe axis with the path normal direction. During execution, the robot followed the pre-planned paths, while the gauge probe maintained contact with the workpiece surface and transmitted measurement data to the host computer for recording.

Six different paths were planned and tested on the curved surface, and the corresponding normal deviations are shown in Fig. 14. The results indicate that all measured deviations were within the range of -1 mm to 1 mm,

demonstrating that the proposed method can maintain stable normal accuracy on freeform surfaces and meet the requirements of typical industrial applications.

For comparison, manual teaching was also conducted on the same workpiece to generate reference paths. The maximum normal deviation of manually taught paths was measured to be approximately 2–3 mm, which is higher than that of the proposed method. Due to differences in path generation strategies, a direct one-to-one comparison under identical conditions is not feasible; therefore, only an order-of-magnitude comparison is presented. Nevertheless, the comparison indicates that the proposed method exhibits superior normal accuracy and consistency, demonstrating its advantages in both precision and robustness over traditional manual teaching.

It should be noted that the above experimental results are inevitably influenced by both hardware and methodological factors. Hardware-related errors mainly stem from the accuracy of the 3D camera under different working distances and reflectivity, the robot's positioning accuracy, and the compliance of the path-marking tool when contacting freeform surfaces. Method-related errors are associated with point cloud sampling, surface normal estimation, and mesh parameterization, where local deformation is unavoidable in highly curved regions. In addition, hand-eye calibration and tool center point calibration may also introduce deviations. Although these factors cannot be completely eliminated, their influence was minimized through repeated calibration and stable experimental settings, ensuring that the obtained results are representative and reliable. Nevertheless, the proposed method still demonstrates stable accuracy and robustness compared with existing approaches.

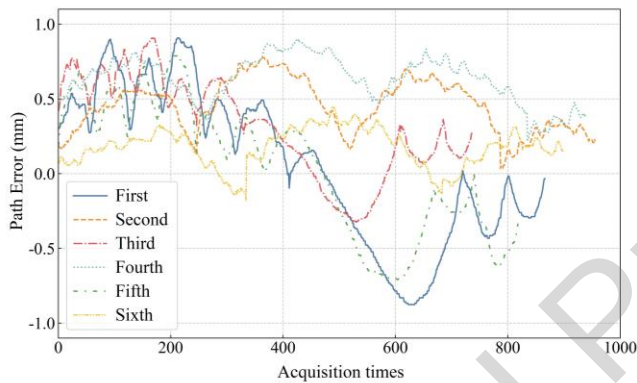


Fig. 14 Normal errors of the path measured in six experiments

5 Conclusions

In this study, a path planning method based on 3D point cloud data and LSCM is proposed, which effectively improves the robot's path planning efficiency and adaptability in complex free-form surfaces and non-standard workpiece grinding. The superiority of the method in high curvature regions is experimentally verified, and the influence of error sources such as point cloud accuracy, image processing and robot execution accuracy on the experimental data is reduced through multiple experiments. Moreover, the normal accuracy of the planned paths on freeform surfaces is evaluated, and all measured deviations are found to be within -1 mm to 1 mm, demonstrating stable performance of the proposed method. Future work will focus on further improving the data acquisition accuracy, optimizing the image processing algorithm and execution accuracy to enhance the accuracy and consistency of path planning.

Funding

Supported by National Natural Science Foundation of China (Grant No. 52175055).

Acknowledgements

The authors sincerely thank Lei Mao for his valuable discussions and suggestions.

CRedit authorship contribution statement

Huimin Shen and Kangming Liu discussed the manuscript; Kangming Liu wrote the manuscript; Huimin Shen reviewed and revised the manuscript; Shuangyuan Wang guided the technical schemes. All authors read and approved the final manuscript.

Declaration of Competing Interest

The authors declare no competing financial interests.

References

- [1] H L Xie, J R Li, Z Y Liao, et al. A robotic belt grinding approach based on easy-to-grind region partitioning. *Journal of Manufacturing Processes*, 2020, 56: 830–844.
- [2] K W Ma, L Han, X X Sun, et al. A path planning method of robotic belt grinding for workpieces with complex surfaces. *IEEE/ASME Transactions on Mechatronics*, 2020, 25: 728–738.
- [3] X Zeng, G Y Zhu, Z H Gao, et al. Surface polishing by industrial robots: A review. *The International Journal of Advanced Manufacturing Technology*, 2023, 125: 3981–4012.
- [4] X L Ke, Y H Yu, K S Li, et al. Review on robot-assisted polishing: status and future trends. *Robotics and Computer-Integrated Manufacturing*, 2023, 80: 102482.
- [5] P Xu, C F Cheung, C J Wang, et al. Novel hybrid robot and its processes for precision polishing of freeform surfaces. *Precision Engineering*, 2020, 64: 53–62.
- [6] Q Y Wang, Y C Cheng, W H Jiao, et al. Virtual reality human-robot collaborative welding: A case study of weaving gas tungsten arc welding. *Journal of Manufacturing Processes*, 2019, 48: 210–217.
- [7] Y Wang, X W Wang, S Y Chen, et al. Multi-station multi-robot welding system planning and scheduling based on stnsga-d: An industrial case study. *IEEE Transactions on Automation Science and Engineering*, 2024, 21: 7465–7479.
- [8] Z S Wang, Y H Shi, S Y Jiang. Passive vision based three-dimensional seam tracking using a stereo vision system with a single HDR camera. *Measurement*, 2025, 242: 116106.
- [9] Y Wang, L X Xie, H Wang, et al. Intelligent spraying robot for building walls with mobility and perception. *Automation in Construction*, 2022, 139: 104270.
- [10] W Chen, J J Liu, Y Tang, et al. Automatic spray trajectory optimization on bézier surface. *Electronics*, 2019, 8(2): 168.
- [11] M Wang, Y M Song, P F Wang, et al. Grinding/cutting technology and equipment of multi-scale casting parts. *Chinese Journal of Mechanical Engineering*, 2022, 35: 97.
- [12] J Alqallaf, J A Teixeira. Blade roughness effects on compressor and engine performance—a CFD and thermodynamic study. *Aerospace*, 2021, 8(11): 330.
- [13] Y Zhang, H D Liu, W K Cheng, et al. A novel trajectory planning method for robotic deburring of automotive castings considering adaptive weights. *Robotics and Computer-Integrated Manufacturing*, 2024, 86: 102677.
- [14] F Xu, B Zi, J Y Wang, et al. Multi-objective trajectory optimization for rigid-flexible coupling spray-painting robot integrated with coating process constraints. *Chinese Journal of Mechanical Engineering*, 2024, 37: 152.
- [15] J C Jiao, W Tian, L Zhang, et al. Variable stiffness identification and configuration optimization of industrial robots for machining tasks. *Chinese Journal of Mechanical Engineering*, 2022, 35: 115.
- [16] S N Bastida, C Y Lin. Autonomous trajectory planning for spray painting on complex surfaces based on a point cloud model. *Sensors*, 2023, 23(24): 9634.
- [17] M Hosseinzadeh, R Himat, Z M Sahihazar, et al. Probabilistic health risk assessment of occupational exposure to BTEX in a paint manufacturing plant using monte-carlo simulation. *Human and Ecological Risk Assessment: An International Journal*, 29(3–4): 859–880.
- [18] W L Li, G Wang, G Zhang, et al. Interference-free inspection path generation for impeller blades using an on-machine probe. *IEEE/ASME Transactions on Mechatronics*, 2017, 22(3): 1218–1226.
- [19] W Z Zhang, W W Huang, Y J Shen, et al. Adaptive five-axis scanning of deformed surfaces based on real-time path planning. *IEEE Transactions on Instrumentation and Measurement*, 2023, 72: 7503313.
- [20] J Jahanpour, M Motallebi, M Porghoveh. A novel trajectory planning scheme for parallel machining robots enhanced with NURBS curves. *Journal of Intelligent & Robotic Systems*, 2016, 82: 257–275.
- [21] X M Zhu, P Martensson, L Hanson, et al. Automated assembly quality inspection by deep learning with 2D and 3D synthetic CAD data. *Journal of Intelligent Manufacturing*, 2025, 36(4): 2567–2582.
- [22] W H Zhang, Z H Cai, T Y Chen, et al. A path planning method for polishing thermal barrier coatings on the surface of aero-engine blades based on point cloud slicing. *Journal of Thermal Spray Technology*, 2024, 33: 33–50.
- [23] Z Q Zhu, Z T Chen, Y Zhang, et al. A novel model reconstruction strategy for polishing turbine blade based on position optimization of sectional curve. *The International Journal of Advanced Manufacturing Technology*, 2022, 119: 5011–5023.
- [24] F Yang, Z H Cai, Y P Chen, et al. A robotic polishing trajectory planning method combining reverse engineering and finite element mesh technology for aero-engine turbine blade tbc. *Journal of Thermal Spray Technology*, 2022, 31: 2050–2067.
- [25] W T Xie, D N Song, W C Tang, et al. Auxiliary support path planning for robot-assisted machining of thin-walled parts with non-uniform thickness and closed cross-section based on a neutral surface. *Journal of Manufacturing Processes*, 2025, 147: 16–28.
- [26] B Lévy, S Petitjean, N Ray, et al. Least squares conformal maps for automatic texture atlas generation. *ACM Transactions on graphics*, 2002, 21: 362–371.
- [27] W Li, Z Yin, Y Huang, et al. Tool-path generation based on angle-based flattening. *Proceedings of the Institution of Mechanical Engineers, Part B: Journal of Engineering Manufacture*, 2010, 224(10): 1503–1509.
- [28] J Liu, D Bai, L Chen. 3-d point cloud registration algorithm based on greedy projection triangulation. *Applied Sciences*, 2018, 8(10): 1776.
- [29] Floater M S, Hormann K. Surface parameterization: A tutorial and survey. *Advances in Multiresolution for Geometric Modelling*, 2005: 157–186.
- [30] K Hormann, B Lévy, A Sheffer. Mesh parameterization: Theory and practice. *ACM SIGGRAPH ASIA 2008 Courses*, December 10, Singapore, 2008: 1–87.

Authors' Information

Huimin Shen, received the Ph.D. degree from *School of Mechanical Engineering, Zhejiang University, China*, in 2015. She is a professor at *School of Mechanical Engineering, University of Shanghai for Science and Technology (USST), China*. Her main research interests include mechanism and robotics.

Kangming Liu, received his master degree in mechanical engineering from *University of Shanghai for Science and Technology, China*, in 2025. His research interests include the field of robot control and trajectory planning, with a particular focus on vision-based robot trajectory planning.

Shuangyuan Wang, received the Ph.D. degree from *School of Mechanical Engineering, Shanghai Jiao Tong University, China*, in 2017. He is an associate professor at *School of Mechanical Engineering, University of Shanghai for Science and Technology (USST), China*. His main research interests include smart manufacturing and machine learning.